

Pritesh Gurjar

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EDUCATION

Case Western Reserve University

Master of Science in Computer Science | GPA: 3.76/4.0

Cleveland, OH

Aug 2024 – May 2026

- Relevant Coursework:** Generative AI, Distributed Systems, High-Performance Computing, Machine Learning, Advanced Algorithms

TECHNICAL SKILLS

ML/GenAI Engineering: RAG Pipelines, Semantic Search, Re-ranking (Cohere), Prompt Engineering, Hallucination Evaluation, Vector Embeddings (Milvus/FAISS)

Agentic AI & Algorithms: LangGraph, AutoGen, Model Context Protocol (MCP), Tool Calling Architectures, Multi-Agent Systems, NDCG/MRR Evaluation

Programming & Frameworks: Python (FastAPI, PySpark), Java (Spring Boot), React.js, TypeScript, RESTful APIs

Infrastructure & Optimization: Docker, Kubernetes, Azure, Linux/Unix, CI/CD Pipelines, High-Performance Computing (HPC)

EXPERIENCE

AI Research Assistant – Agentic AI Systems Architect

University Technology (UTech), CWRU

Oct 2024 – Present

Cleveland, OH

- Architected a **multi-agent system** using **LangGraph** and **Model Context Protocol (MCP)** to orchestrate specialized agents, enabling autonomous research workflows for 500+ faculty members.
- Designed and implemented secure **tool-calling architectures** allowing LLMs to interface with internal APIs while strictly maintaining FERPA data compliance.
- Built an evaluation framework utilizing **NDCG and MRR metrics** to benchmark retrieval quality, resulting in a 20% improvement in answer relevance through iterative re-ranking optimizations.
- Standardized reusable AI components and prompt caching mechanisms on Linux servers, reducing token costs by 30% and establishing a robust baseline for future agent deployments.

Software Engineer – Full stack & Data Engineering

NielsenIQ

Jul 2022 – Dec 2024

Pune, India

- Engineered scalable full-stack solutions using **Java Spring Boot** and **React**, optimizing RESTful interfaces to support high-throughput client applications with 99.9% uptime.
- Designed and implemented high-volume data pipelines using **PySpark** and **NoSQL database models (MongoDB)**, ensuring high-quality data availability for downstream ML feature stores processing 10M+ records.
- Optimized web server configuration using **Nginx** and cloud load balancers, reducing critical API latency by 60% through efficient thread management and asynchronous processing.
- Deployed microservices on **Docker and Kubernetes**, enabling cross-functional teams to reuse data ingestion modules and reducing code duplication by 40% across the infrastructure.

PROJECTS

EduAssist-AI – Full-Stack Multimodal RAG Pipeline | *React, FastAPI, Whisper, CLIP, Vector DB*

2024

- Architected a full-stack GenAI application orchestrating **Whisper v3** (audio processing), **CLIP** (vision encoding), and **T5** (text generation) to convert unstructured lecture videos into structured knowledge graphs.
- Implemented a RAG-based retrieval system using **FAISS** and **FastAPI**, enabling semantic search over multimodal content with sub-200ms latency.
- Designed the frontend interface for visualization and retrieval, achieving 92% accuracy in context relevance by fine-tuning T5 on domain-specific educational datasets.

TestPilot Auto – Agentic Code Generation System | *Python, Selenium, AST, LLM*

2024

- Built an **agentic workflow** leveraging Code Llama that autonomously analyzes DOM structures and generates robust Selenium/Cypress test scripts, reducing manual authoring time by 70%.
- Implemented **iterative self-correction capabilities** allowing the agent to execute generated tests in a sandbox, parse error logs, and autonomously refine code failures.
- Integrated advanced context management and prompt engineering, achieving 85% code syntax accuracy and lowering test maintenance overhead by 60%.

Library AI Search Agent – Enterprise Retrieval System | *RAG, NoSQL, NLP*

2024

- Engineered an intelligent retrieval agent connecting to catalog databases, processing over 10,000 queries with a custom ranking algorithm that prioritized academic relevance and context proximity.
- Optimized the **vector search retrieval** pipeline to handle complex natural language queries efficiently, resulting in a 45% decrease in librarian support tickets related to resource discovery.

ACHIEVEMENTS

Published Researcher: IRJET - Traffic Sign Recognition using Deep Learning (Achieved 97% accuracy in classification)

IBM Gold Winner: Recognized for excellence in Security Technologies Research (2020)

Graduate Teaching Assistant: Served as TA for Data Structures and Algorithms course (C++)